

Alok Kumar Mishra

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EDUCATION

- Indian Institute of Information Technology, Jabalpur** Madhya Pradesh, India
Bachelor of Technology - Electronics & Communication Engineering Aug 2024 – Present

EXPERIENCE

- DigiLocker (MeitY), Government of India** Remote
Security Contributor (AppSec) Feb 2025 – Apr 2025
 - Application Security & Android RE:** Responsibly disclosed **10+ P1/P2 vulnerabilities** in DigiLocker’s Android app (**500M+ users**) via dynamic/static analysis using **Frida, JEB, Radare2**.
 - Remediation & Recognition:** Worked with DigiLocker engineers on fixes through detailed reports; received **official Govt. of India acknowledgment** for responsible disclosure.
- Rizin Project (Open-Source RE Framework)** Remote
Core Contributor – RE Tooling Nov 2025 – Dec 2025
 - Binary Hardening:** Improved ELF parsing robustness against malformed binaries and DoS vectors using bounds checks aligned with **CVE-style hardening**.
 - Core Tooling:** Implemented De Bruijn decoding shortcuts and unified address APIs across core modules; added regression test binaries and RE documentation.

PROJECTS

- RzWeb – Rizin on Web (WebAssembly Reverse Engineering Platform)** GitHub
WebAssembly (WASM), WASI, TypeScript, Python, Shell, Emscripten
 - Built **RzWeb**, a browser-based reverse engineering IDE by compiling the **Rizin** toolchain to **WebAssembly (WASM)** using WASI/Emscripten, enabling native-grade analysis fully in-browser.
 - Engineered **RzWasi** integration to run Rizin modules inside a WASI runtime, solving syscall/runtime constraints via host bindings and sandbox-safe execution.
 - Implemented core RE workflows (disassembly, interactive console, graph-based navigation) with a responsive TypeScript UI, optimized for low-latency iteration on binaries.
 - Shipped a production demo at *rizin.pages.dev* with reproducible build scripts and deployment automation, reducing setup friction from hours to **under 2 minutes**.
- Arthyx – Quant Finance Document Intelligence** GitHub
Next.js, TypeScript, Neo4j, Vector Search
 - Engineered a **RAG-based quant system** for deep analysis of **500+ page financial PDFs** (Basel III, NPA, CRAR).
 - Modeled financial entities and risk propagation using **Neo4j knowledge graphs**, reducing manual review by **40%**.
- HexaCiphers – Social Media Intelligence & OSINT** GitHub
React, Python, NLP
 - Built a real-time multilingual OSINT platform for Twitter/X campaign and narrative analysis.
 - Implemented bot detection and sentiment pipelines achieving **80% accuracy** for cyber threat intelligence use cases.

TECHNICAL SKILLS

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, Bash, SQL, Smali, HTML/CSS
- Cybersecurity / RE:** VAPT, Bug Bounty, AppSec, Android RE, Frida, Ghidra, IDA Pro, JEB, Radare2, Burp, Metasploit
- Web / Backend:** React, Next.js, Node.js, FastAPI, Firebase, Tailwind, REST APIs
- AI / Finance:** RAG Systems, NLP, Vector Search, Neo4j, Gemini API, Basel III, NPA/CRAR
- Tools:** Git, Docker, Linux, Android Studio, Postman, Vercel

ACHIEVEMENTS

- Acknowledged **Security Contributor** by DigiLocker (GOI) for 10+ critical vulnerabilities (500M+ users).
- 1st Place** – Can You Hack It Hackathon (200+ students), Team Lead.
- Top 20 / 700+ teams** – Hack To Future 2025, IIT Gandhinagar.
- JEE Main Top 1.2%;** CBSE Merit (**100/100 CS**); **Global Rank 385** in CodeChef Starters 176C.

VOLUNTEER EXPERIENCE

- Jagrati – Non-Profit Education Initiative** Jabalpur, India
Mathematics Tutor (Volunteer) Aug 2024 – Present
 - STEM Education Outreach:** Providing free mathematics tutoring to underprivileged students preparing for Navodaya entrance examinations; designed simplified study plans to improve conceptual understanding.
- TechFest “AbhiKalpan” – IIITDM Jabalpur** Jabalpur, India
Student Volunteer Apr 2025
 - Event Operations & Coordination:** Assisted in planning and execution of the institute’s annual technical festival; managed logistics and on-site coordination for the Laser Constellation event, ensuring smooth participant flow.